PC 23CS201 Data Structures **3 0 0 3**

At the end of the course, students will be able to

Code	Course Outcomes	Mapping with POs / PSOs					DoK
		PO1	PO2	PO3	PSO1	PSO2	
23CS201.1	Illustrate the role of linear data structures in organizing and accessing data efficiently in algorithms	3	3	2	3	2	L1 – L3
23CS201.2	Analyze linked lists for dynamic data storage, demonstrating understanding of memory allocation.	3	3	2	3	2	L1 – L4
23CS201.3	Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems	3	3	3	3	2	L1 – L3
23CS201.4	Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between deques and priority queues, and apply them appropriately to solve data management challenges	3	3	3	3	2	L1 – L4
23CS201.5	Identify scenarios where hashing is advantageous and develop hash-based solutions for specific problems	3	3	3	3	2	L1 – L4

All the COs are mapped to PO12 as few self-learning topics are inbuilt in syllabus promoting autonomous learning

Unit I: Introduction to Data Structures **9 Hours**

Types of Data Structures, Definition and importance of linear data structures, Abstract data types (ADTs) and their implementation, Importance of Non-Linear Data Structures, Overview of time and space complexity analysis for linear data structures. Searching Techniques: Linear & Binary Search. Sorting Techniques: Bubble sort, Selection sort, Insertion Sort

COs: CO1

Self - Learning Topic: Analyze Time Complexity of Binary search

Unit II: Linked Lists **9 Hours**

Singly linked lists: representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.

COs: CO2

Self - Learning Topic: Swap nodes in a Linked List (without Swapping data)

Unit III: Stacks **9 Hours**

Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list etc.

COs: CO3

Self - Learning Topic: Types of stacks

Unit IV: Queues & Deques **9 Hours**

Queues: Introduction to queues: properties and operations, implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.

Deques: Introduction to deques (double-ended queues), Operations on deques and their applications.

COs: CO4

Self - Learning Topic: Different types of Queues

Unit V: Trees, Hashing **9 Hours**

Trees: Introduction to Trees, Binary Search Tree – Insertion, Deletion & Traversals.

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc, Applications of Data Structures **COs: CO5**

Self - Learning Topics: Types of Binary Trees

Board of Studies		Computer Science and Engineering	
Approved in: BoS No. VI		October 06, 2023	
Approved in: ACM No. VIII		October 21, 2023	
Expert talk (To be delivered by SMEs from industries)		**COs**	**POs**
1	Graphs	CO4	PO1, PO2, PO3
2	AVL Tree, B Tree, B+ Tree	CO5	PO1, PO2, PO3

Text Books

1. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", 2nd Edition, Pearson Education, 2002
2. Ellis Horowitz, Sartaj Sahni, "Fundamentals of Data Structures in C", 2nd Edition, Universities Press, 2014

Reference Books

1. Kurt Mehlhorn and Peter Sanders, "Algorithms and Data Structures: The Basic Toolbox", Springer, 2008
2. Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft, "Data Structures and Algorithms", 1st Edition, Pearson Education India, 2002
3. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, "Introduction to Algorithms", PHI Learning Pvt. Ltd., 3rd Edition, 2010
4. Robert Sedgewick, "Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms", Addison-Wesley; 3rd Edition, 2001
5. Reema Thareja, "Data Structures Using C", 2nd Edition, Oxford Publication, 2014

Web References

1. https://github.com/topics/data-structures-c
2. https://www.hackerrank.com/domains/data-structures
3. http://cslibrary.stanford.edu/
4. https://www.tutorialspoint.com/dsa_using_c/dsa_using_c_quick_guide.html

Internal Assessment Pattern

Cognitive Level	Internal Assessment #1 (%)	Internal Assessment #2 (%)
L1	15	15
L2	35	35
L3	40	40
L4	10	10
Total (%)	100	100

Sample Short and Long Answer Questions of Various Cognitive Levels

L1: Remember

1. List any five operations in single linked list
2. What is the principle of stack?
3. What are the operations of queue?
4. Define single linked list with suitable example
5. List any two linear data structure